Finance theory and modeling

Lecture 6

Basic Options Theory

(Chapter 12, Luenberger)

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Options

- An **option** is the right, but not the obligation to buy (**call** option), or sell (**put** option) an asset under specified terms.
- These include the date (**expiration/maturity**)
- The price for which you buy/sell (**strike price**)
- And whether or not you can use the option before the expiration data (**American** option) or cannot use it before expiration (**European** option)
- You pay a price for such an option, it is often also called the **premium**.

Option example

- You doubt whether you want to buy a house.
- You buy an option on the house.
- So it is a call option.
- This means you do not *have* to buy it.
- The price you agree on is $200,000.
- You pay a fraction of this for the option, say $15,000.
- You have to decide 1 year from now, afterwards the option expires.

Option example

- If, after 1 year, the house is worth $\$300,000$, you use the option to buy it cheap ($\$200,000$) and gain $\$100,000$.
- If the house is worth $\$150,000$, you will not use your option. (It would not make sense to pay more).
- It is easy to see the value of the option after 1 year; in these two examples, the option either saves you $\$100,000$ or it is useless ($\$0$).
- The question we try to answer: why pay $\$15,000$ now? i.e. what is the value of the option?

Option Quotes

Option Chain for Intel Corp. (INTC)
$ 21.25 -0.61 (-2.79%) **Volume: 57.96 m** **3:23 PM EDT Mar 27, 2008**

Option Filter: Composite | All | At the Money | GO | Help

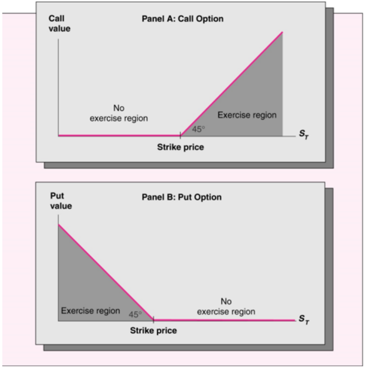
Feb 08 | Mar 08 | **Apr 08** | May 08 | Jul 08 | Oct 08 | Jan 09 | Jan 10 | All

Calls	Last	Chg	Bid	Ask	Vol	Open Int	Strike	Puts	Last	Chg	Bid	Ask	Vol	Open Int
@NQDV	9.19	0.14	8.75	8.85	2	824	12.50	@NQPV	0.02	-		0.02	1	2887
@NQDC	6.80		6.20	6.35	0	2355	15.00	@NQPC	0.02			0.03	0	3290
@NQDQ	5.15		5.20	5.40	0	1654	16.00	@NQPQ	0.01	-0.04	0.01	0.03	10	1537
@NQDW	3.95	-0.50	3.80	3.90	212	7107	17.50	@NQPW	0.07	0.01	0.07	0.09	173	20597
@NQDT	2.54	-0.47	2.50	2.53	345	3953	19.00	@NQPT	0.23	0.07	0.22	0.24	124	13710
@NQDD	1.75	-0.47	1.72	1.74	2527	33593	20.00	@NQPD	0.45	0.15	0.44	0.45	2702	59325
@NQDU	1.10	-0.36	1.06	1.08	4466	23528	21.00	@NQPU	0.76	0.20	0.78	0.79	3405	28891
@NQDX	0.42	-0.23	0.41	0.42	3452	53399	22.50	@NQPX	1.64	0.41	1.63	1.64	1544	40392
@NQDB	0.12	-0.09	0.13	0.14	208	11807	24.00	@NQPB	2.74	0.51	2.82	2.86	31	2943
@INQDE	0.06	-0.04	0.06	0.07	102	57633	25.00	@INQPE	3.69	0.64	3.70	3.85	18	27803
@INQDY	0.02		0.01	0.02	0	34576	27.50	@INQPY	6.16		6.20	6.30	0	4220
@INQDF	0.01	-		0.01	66	30721	30.00	@INQPF	9.00		8.70	8.80	0	361

Payoffs for options

- The payoff of an option at expiration is similar to the grading of this course:
- Suppose we have strike price K and Asset price S at expiration.
- For a Call: $\mathrm{C} = \max(0, S - K)$
- For a Put: $\mathrm{P} = \max(0, K - S)$
- Again, the value at expiration is easy, but what to pay for the option right now?

In the money



- You can compare the strike price with the (current) price of the underlying asset.
- **In the money** means you get a positive payoff.
- **At the money** means the strike price is equal to the price of the asset.
- **Out of the money** means you will not exercise (and get nothing).

Which options are in the money?

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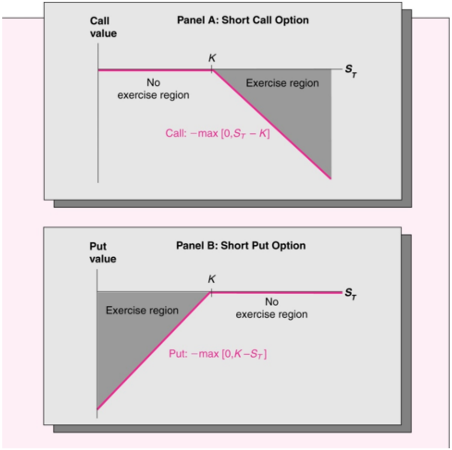
Shorting options



Q: If you can only loose money, why short at all?

Put Call-Parity

- We can derive the price for a Call option, given an otherwise identical Put option...
- At expiration T:
  1. $C_T = \max(0, S_T - K_T)$
  2. $P_T = \max(0, K_T - S_T)$

$$C_T - P_T = \max(0, S_T - K_T) - \max(0, K_T - S_T) = S_T - K_T$$

- And so for today:

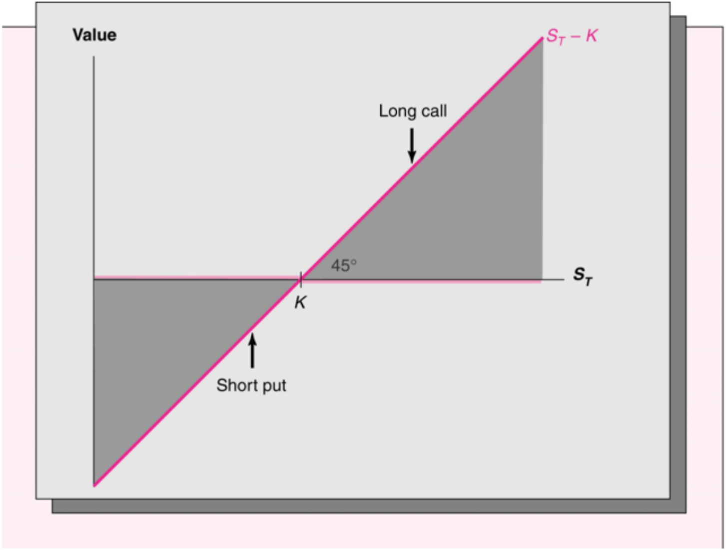
$$C_t - P_t = S_t - K_t = S_t - PV(K_T)$$

Explanation PC parity

Values at Expiration:

	If $S_T < K$	If $S_T \geq K$
long call:	0	$S_T - K$
short call:	0	-$(S_T - K)$
long put:	$K - S_T$	0
short put:	-$(K - S_T)$	0

Explanation PC parity



PC parity

Investment	*Cost of Acquiring the Investment Today (1)*	*Cash Flows at the Expiration Date if* $\tilde{S}_T < K$ *(2)*	*Cash Flows at the Expiration Date if* $\tilde{S}_T > K$ *(3)*
Investment 1:			
Buy call and write put	$c_0 - p_0$	$\tilde{S}_T - K$	$\tilde{S}_T - K$
= Long position in a call	$= c_0$	$= 0$	$= \tilde{S}_T - K$
and short position in a put	$- p_0$	$-(K - \tilde{S}_T)$	-0
Investment 2:			
Tracking portfolio	$S_0 - PV(K)$	$\tilde{S}_T - K$	$\tilde{S}_T - K$
= Buy stock	$= S_0$	$= \tilde{S}_T$	$= \tilde{S}_T$
and borrow present value of K	$-PV(K)$	$-K$	$-K$

Yields put-call parity $p_0 + S_0 - c_0 = PV(K)$.

PC parity with numbers

Investment	*Cost of Acquiring the Investment Today (1)*	**Cash Flows at the Expiration Date if S at that time is:** *\$45 (2)*	*\$52 (3)*	*\$59 (4)*
Investment 1:				
Buy call and write put	$c_0 - p_0$	$-\$5$	\$2	\$9
= Long position in a call	= c_0	= 0	= (\$52 − \$50)	= (\$59 − \$50)
and short position in a put	$-p_0$	−(\$50 − \$45)	−0	−0
Investment 2:				
Tracking portfolio	$S_0 - PV(50)$	$-\$5$	\$2	\$9
= Buy stock	= S_0	= \$45	= \$52	= \$59
and borrow present value of K	$-PV(\$50)$	−\$50	−\$50	−\$50

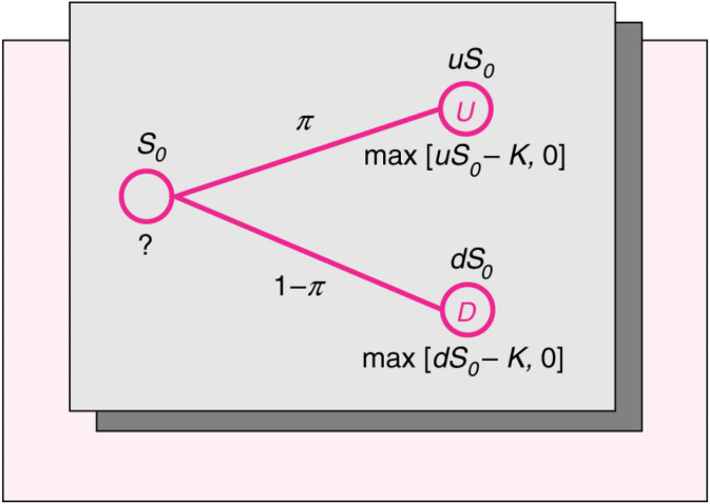
Notes PC parity

- The Put-Call parity:
- Does only hold for European options.
- Does not hold if dividends are paid before expiration.
- Still can be useful for American options.

Relationship of value with expiring date

- European Options: put-call parity.
- American Options:
  1. Call
     - With dividend pay.
     - Without dividend pay.
  2. Put options

Valuing options in isolation



Risk-neutral probabilities

- Assume 10% interest rate. Say stock value is 50 now and next period 40 or 60.

$$\frac{\pi(60-50)+(1-\pi)(40-50)}{50}=0.1 \Rightarrow$$
$$\frac{10\pi+10\pi-10}{50}=0.1$$

- $\pi=0.75$ is risk neutral probability.

Risk-neutral probabilities

Different ways of looking at it.

1. "Mathematical Trick" that avoids calculating the portfolio replication strategy...
2. We know from chapter 9:

$$P_i = \frac{E[U'(X^*)X_i]}{E[U'(X^*)]R_f}$$

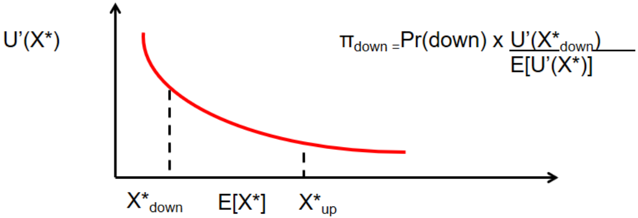
For two outcomes (up, down) this is:

$$\begin{aligned} P_i &= Pr(up) \cdot \frac{U'(X^*_{up})X_{i,up}}{E[U'(X^*)]R_f} + Pr(down) \cdot \frac{U'(X^*_{down})X_{i,down}}{E[U'(X^*)]R_f} \\ &= \frac{(\pi_{up}X_{i,up} + \pi_{down}X_{i,down})}{R_f}, \end{aligned}$$

with $\pi_{up} = Pr(up) \cdot \frac{U'(X^*_{up})}{E[U'(X*)]}$.

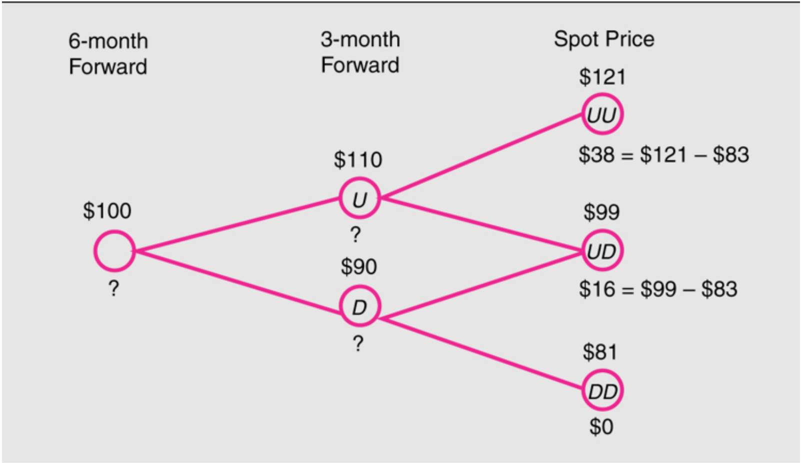
Risk-neutral probabilities

Marginal utility changes real probabilities into risk-neutral probabilities in an intuitive way...



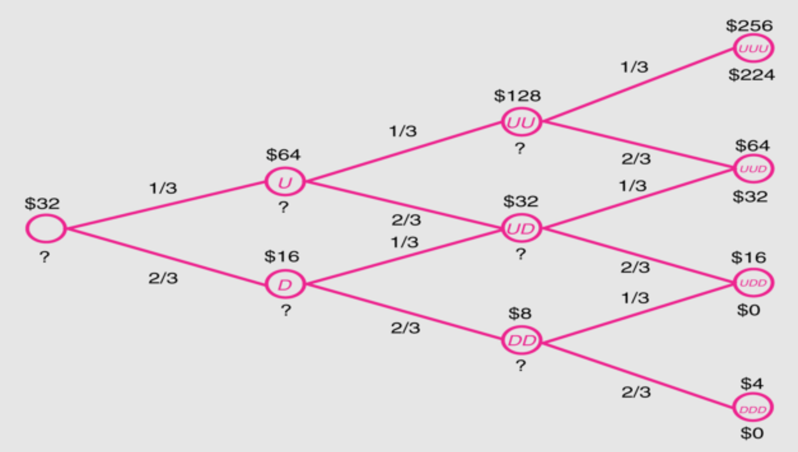
"A risk-averse person behaves the same as a risk-neutral person that attaches relatively more probability to "bad" events..."

Make tree larger...



Even larger trees

You can increase the number of periods.



American puts

- For american puts it is sometimes beneficial to exercise the put earlier.
- You have to take this into account when evaluating the nodes in the tree.
- If continuation value is lower compared to exercising immediately, you will of course exercise earlier.
- Intuition: what if $S = 0$? Can you get even more in the Money?

General risk-neutral pricing

- The method does not only work for options.
- As long as we know the payoff of the derivative we want to price at every node in the tree, we can value the derivative using the same principal. $\Rightarrow$ Expectation $\hat{E}$ of payoffs.

Just use risk neutral probabilities and apply:

$$P = \hat{E}\Big[\sum d_k f_k\Big],$$

where d_k is the risk-free discount factor in node k and f_k the payoff at node k.

If the risk-free rate does not depend on k:

$$P = \frac{1}{R} \cdot \hat{E}\Big[\sum f_k\Big],$$

e.g. for a Call:

$$C = \frac{1}{R} \cdot \hat{E}\big[\max(S_T - K, 0)\big]$$

Useful in practice?

- Very.
- The *absolute* pricing of assets (equity, real estate), using factor models/CAPM etc. is not very accurate.
- The *relative* pricing of derivatives (options, futures, turbos) is quite accurate.
- It assumes no arbitrage, and since we know underlying prices, we can almost perfectly replicate payoffs.

Assignment 2

- **Option pricing**
  Deadline: 25th of October 2024, 9.00 pm
- Hand in hardcopy either during tutorial or in my mailbox. 3 pages is max. Every extra page is a point reduction.
- **Group size: 3 or 4 students**
  In this assignment, you will study European and American call and put options.

Assignment 2

- You are free to use any software package.
- E.g. Python, R, Excel, Matlab, etc.
- It may be useful to have a solver routine available (this is also possible in Excel).
- Otherwise you may have to do a lot of "trial and error" since explicit solutions are not always possible.